



1

The Challenge

Students are asked to:

- Develop a conceptual AI/Data Science solution.
- Work with simulated accounting datasets.
- Deliver Executive Summary + Mock-up Presentation.
- Goal: Demonstrate improvements in processes and performance.



2

Positioning of the Work

Purpose:

- Build demonstrative prototypes with simulated data.
- Apply AI/Data Science to accounting.

Deliverables:

- Executive Summary, Mock-up, Evidence of benefits.

Excluded:

- ERP integration, Enterprise RPA, Industrial compliance.
- Objective: Feasible prototypes at Bachelor level.

3

Business Cases Index

1. Smart Reconciliation
2. Cash Flow Crystal Ball
3. Fraud Radar
4. Fair Share
5. Margin Watch
6. Closing Guardian
7. Cost Foresight
8. Compliance Copilot
9. Inventory Brain
10. Green Ledger

4

Smart Reconciliation – Multi-Entity Accounting in the Food Industry

Industry Focus	Food manufacturing and distribution
Problem	Multinational food groups face difficulties reconciling accounts across subsidiaries and currencies. Differences in exchange rates, reporting standards, and ERP systems cause inconsistencies and delays.
Application	Prototype AI model on simulated datasets reconciles intercompany transactions, standardizes exchange rates, and flags anomalies. This reduces closing time and improves transparency.
Deliverables	Executive Summary, dashboard mock-up, explanation of benefits.
Academic Scope	Feasible with Python (pandas, anomaly detection). Excluded: ERP integration, real multi-currency reconciliation.
Accounting Focus	Apply horizontal and vertical balance sheet analysis. Use Current Ratio and Working Capital to assess liquidity consistency across entities. Identify and reconcile variances in Accounts Receivable Turnover.

5

Cash Flow Crystal Ball – Predictive Liquidity in Construction

Industry Focus	Construction
Problem	Construction firms face liquidity issues when inflows/outflows are not forecast correctly.
Application	Predictive model on simulated data (invoices, payables, recurring payments) forecasts daily cash flow.
Deliverables	Executive Summary, forecasting prototype, benefits analysis.
Academic Scope	Regression or ARIMA feasible. Excluded: real banking integrations.
Accounting Focus	Use Operating Cash Flow Ratio, Days' Sales Uncollected, and Days' Payables Outstanding. Link predictions to Liquidity and Efficiency Ratios for project-level cash management.

6

Fraud Radar – Billing Anomaly Detection in Energy Services

Industry Focus	Energy services
Problem	Fraudulent activities such as duplicate invoices, fake suppliers, or abnormal amounts often unnoticed.
Application	Prototype anomaly detection system analyzes simulated invoice datasets for suspicious transactions.
Deliverables	Executive Summary, fraud detection dashboard mock-up.
Academic Scope	Clustering, isolation forest feasible. Excluded: enterprise audit integration.
Accounting Focus	Apply Profit Margin and Revenue Growth analysis via trend and vertical methods. Detect anomalies using comparative income statements and ratio deviations.

7

Fair Share – Overhead Allocation in Manufacturing

Industry Focus	Manufacturing
Problem	Indirect costs like IT or maintenance allocated arbitrarily, creating conflicts and mistrust.
Application	Prototype model uses simulated production and machine usage data to allocate costs fairly.
Deliverables	Executive Summary, cost allocation model mock-up, fairness evidence.
Academic Scope	Regression/clustering feasible. Excluded: integration with advanced controlling systems.
Accounting Focus	Compute Gross Margin Ratio and Return on Total Assets (ROA) before/after allocation. Use DuPont Analysis to decompose profitability impacts of overhead redistribution.

8

Margin Watch – Real-Time Profitability in Retail

Industry Focus	Retail
Problem	Retail margins are thin and fluctuate; analysis comes too late.
Application	Prototype dashboard with simulated cost/revenue data monitors margins in near real time.
Deliverables	Executive Summary, dashboard mock-up, decision-making improvements.
Academic Scope	Streamlit/Power BI feasible. Excluded: enterprise-scale retail deployment.
Accounting Focus	Analyze Profit Margin, Inventory Turnover, and Days' Sales in Inventory. Perform common-size income statement comparisons across product categories.

9

Closing Guardian – Error Prevention in Logistics Closing

Industry Focus	Logistics and transport
Problem	Monthly/quarterly closings are manual, slow, and error-prone.
Application	Prototype with Python scripts checks simulated closing data for inconsistencies and suggests corrections.
Deliverables	Executive Summary, script demonstration, efficiency analysis.
Academic Scope	Python validation scripts feasible. Excluded: professional RPA platforms.
Accounting Focus	Evaluate Working Capital Cycle, Asset Turnover, and Debt Ratio. Use comparative balance sheets to validate accuracy of accruals and provisions.

10

Cost Foresight – Forecasting Production Costs in Steel

Industry Focus	Steel manufacturing
Problem	Volatile energy/raw material costs make production costs unpredictable.
Application	Predictive model integrates simulated accounting data with market indices to forecast costs.
Deliverables	Executive Summary, forecasting model mock-up, benefits analysis.
Academic Scope	Regression/time series feasible. Excluded: industrial-scale simulations.
Accounting Focus	Apply Gross Margin Trend Analysis and COGS Ratio by cost center. Integrate Inventory Turnover to connect production cost forecasts with asset use.

11

Compliance Copilot – Automated Checks in Banking

Industry Focus	Banking
Problem	Continuous regulatory changes require constant manual checks, increasing compliance risk.
Application	Prototype NLP model analyzes simplified regulatory texts and simulated accounting data to generate alerts.
Deliverables	Executive Summary, NLP model mock-up, compliance benefits.
Academic Scope	Basic NLP feasible. Excluded: full fiscal compliance systems.
Accounting Focus	Include Debt Ratio, Equity Ratio, and Times Interest Earned as compliance metrics. Model must flag breaches of solvency/leverage limits using accounting ratios.

12

Inventory Brain – Predictive Stock Optimization in Automotive Components

Industry Focus	Automotive components manufacturing and spare parts supply chain.
Problem	Inventory levels fluctuate due to inaccurate demand forecasting, creating stockouts or overstocking that increase costs and disrupt production.
Application	Develop a predictive model using machine learning to forecast part-level demand and production scheduling optimization. Simulate inventory data, production scheduling and design a dashboard showing reorder suggestions and risk indicators for critical items.
Deliverable	Executive summary, dataset simulation (production and demand data), mock-up dashboard prototype, and report on expected cost reduction.
Academic Scope	Application of predictive analytics and data visualization to optimize inventory and supply chain efficiency. Students apply forecasting models and decision-support techniques.
Accounting Focus	Incorporate Inventory Turnover, Days' Sales in Inventory, and Acid-Test Ratio. Evaluate forecasted stock levels through their effect on Current Assets and Working Capital.

13

Green Ledger – Sustainability Accounting in Agriculture

Industry Focus	Agribusiness and agricultural production companies.
Problem	Lack of integration between accounting data and environmental metrics prevents accurate ESG reporting and sustainability tracking.
Application	Build a prototype that links accounting transactions with sustainability indicators (CO ₂ emissions, energy, fertilizers). Use AI models to estimate environmental costs and generate automated ESG compliance reports.
Deliverable	Executive summary, simulated ESG dataset, AI-based accounting prototype (mock-up), and sustainability performance analysis.
Academic Scope	Exploration of AI applications in sustainability accounting, data integration, and automated ESG reporting frameworks. Students demonstrate how to quantify and report environmental impact through data science.
Accounting Focus	Develop an ESG-adjusted ROE combining Return on Equity and Profit Margin. Apply Vertical Analysis on ESG-modified income statements to assess sustainable profitability.

14

Key Takeaways & Conclusion

- AI and Data Science can support accounting.
- Focus on demonstrative prototypes with simulated data.
- Deliverables: Executive Summary, mock-up, evidence of benefits.
- Excluded: ERP, RPA, full compliance systems.
- Goal: Conceptual solutions achievable at Bachelor level.

